

Air Quality and Health Impact Analysis: A Machine Learning Approach

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This project aims to predict the health impact of air quality based on various pollutants and weather conditions using machine learning classification models. The dataset includes several air quality metrics, weather data, and health-related metrics that affect the public. Four machine learning models are explored, namely Artificial Neural Networks (ANN), Support Vector Machines (SVM), Decision Trees, and k-Nearest Neighbors (kNN), with ensemble methods applied for better performance.

1 Introduction

Air quality is a major concern for public health worldwide. Poor air quality, caused by pollutants such as particulate matter (PM_{2.5}, PM₁₀), nitrogen dioxide (NO₂), sulfur dioxide (SO₂), and ozone (O₃), has been linked to a wide range of health issues, including respiratory diseases, cardiovascular conditions, and premature death. According to the World Health Organization (WHO), air pollution is responsible for approximately 1.3 million deaths annually, making it one of the leading environmental risk factors for public health. Understanding the relationship between air quality indicators and their health impacts is crucial for mitigating these risks and improving public health.

This project investigates the relationship between air quality indicators and the health impact they have on individuals. The main objective of this project is to use machine learning to classify the health impact based on air quality metrics. The dataset used in this study provides data on air quality, weather conditions, and health-related issues such as respiratory and cardiovascular cases. By utilizing this data, we aim to develop a model that predicts the health impact class.

Recent studies have explored the application of machine learning in predicting air quality and its associated health risks. Gupta et al.[2] in their study "Prediction of Air Quality Index Using Machine Learning Techniques: A Comparative Analysis" compared various machine learning algorithms to predict the Air Quality Index (AQI), highlighting the importance of accurate air quality forecasting. Gore and Deshpande[1], in their work "Prediction of Air Quality Using Machine Learning Models," demonstrated the effectiveness of decision trees and neural networks in predicting air quality levels and associated health risks. Liang et al.[3], in their paper "Predicting Air Quality with Machine Learning Models: A Case Study of Taiwan," employed machine learning algorithms like Gradient Boosting and K-Nearest Neighbors to predict air quality in Taiwan, showing the potential of these methods to inform public health decisions.

This project aims to address the problem of automatic classification of health impacts based on air quality metrics. The primary objective is to evaluate and compare the performance of different machine learning models, including classifiers like Support Vector Machines (SVM), k-Nearest Neighbors (k-NN), and Artificial Neural Networks (ANNs), in predicting health impact classes. Additionally, dimensionality reduction techniques such as Principal Component Analysis (PCA) will be applied to improve model efficiency and reduce overfitting.

This report is organized as follows: Section 2 presents a bibliographic analysis of the relevant literature, highlighting previous research on air quality prediction and its impact on health. Section 3 describes the dataset

used in the analysis, including its structure and key variables. Section 4 outlines the methodology, detailing the preprocessing steps, dimensionality reduction techniques, and machine learning models used. Section 5 presents the experimental results, including performance metrics such as accuracy, error rate, and F1-score. Finally, Section 6 discusses the findings, draws conclusions, and suggests directions for future research.

2 Bibliographic Analysis

The literature on air quality prediction and its health impacts highlights several key approaches, particularly the use of machine learning models to analyze and predict air pollution levels. A growing body of research has explored the application of machine learning algorithms in predicting air quality indices (AQI) and assessing the health risks associated with pollution exposure. Gupta et al. [2], compared various machine learning models, including Random Forest and Support Vector Machines (SVM), to predict AQI. Their study emphasizes how machine learning can significantly improve the accuracy of air quality forecasts, a critical factor for informing public health and environmental policies.

The current literature also pays significant attention to the effectiveness of different machine learning techniques for air quality prediction. Gore and Deshpande [1], in their study demonstrated the successful application of Decision Trees and Neural Networks for forecasting air quality levels. Their findings underline the importance of selecting the right model and configuration to achieve reliable predictions, particularly for real-time air quality monitoring and public health assessments.

Research into air quality prediction has also focused on the application of machine learning to identify the key environmental factors influencing air pollution. Liang et al. [3], applied Gradient Boosting and K-Nearest Neighbors to forecast air quality levels. Their study revealed how machine learning models could not only predict air quality but also highlight the most significant pollutants contributing to poor air quality. This research provides valuable insights into the use of machine learning for air quality management and health risk mitigation.

There is a limited body of literature that integrates air quality data with health impact predictions, and the generalizability of many published studies remains a challenge. While a large volume of research has focused on air quality prediction alone, fewer studies have examined its direct impact on health outcomes. Recent studies, such as those by Gupta et al. [2] and Gore and Deshpande [1], have pointed to the need for models that incorporate both environmental and health data to provide more comprehensive risk assessments.

This study aims to bridge this gap by using machine learning models to classify health impacts based on air quality data. By combining dimensionality reduction techniques, such as Principal Component Analysis (PCA), with classification models, the goal is to improve both the efficiency and accuracy of health impact predictions. The findings of this project will contribute to the growing body of research that explores the integration of air quality and health data for more accurate predictions of public health risks.

3 Dataset Description

The dataset used for this project is the “Air Quality and Health Impact Dataset,” obtained from Kaggle. It contains a total of 5,811 records, each containing data on various air quality indicators, weather conditions, and health impacts. The goal of this project is to predict the health impact class based on these air quality metrics.

The dataset includes the following key attributes:

- **RecordID:** A unique identifier for each data entry.
- **Air Quality Metrics:** These include various air quality indicators such as: AQI (Air Quality Index), PM10 (Particulate Matter 10), PM2.5 (Particulate Matter 2.5), NO2 (Nitrogen Dioxide), SO2 (Sulfur Dioxide), O3 (Ozone).
- **Weather Conditions:** This includes information like: Temperature, Humidity, WindSpeed.

- **Health Impact Metrics:** Includes the following health-related outcomes: Respiratory Cases, Cardiovascular Cases, Hospital Admissions.
- **HealthImpactClass:** The target variable, categorizing health impacts into five classes: Very High, High, Moderate, Low, and Very Low.

This dataset is used for classification tasks, where the target variable HealthImpactClass is predicted based on the air quality and health data.

3.1 Key Characteristics

The dataset has the following key characteristics:

- **Number of entries:** 5,811 entries, each corresponding to a data point with various features such as air quality indicators and health outcomes.
- **Number of classes:** 5 distinct health impact classes.
- **Attributes:** Includes both continuous variables (e.g., PM2.5, CO) and categorical variables (e.g., HealthImpactClass).

The following pie chart illustrates the distribution of the HealthImpactClass across the dataset:

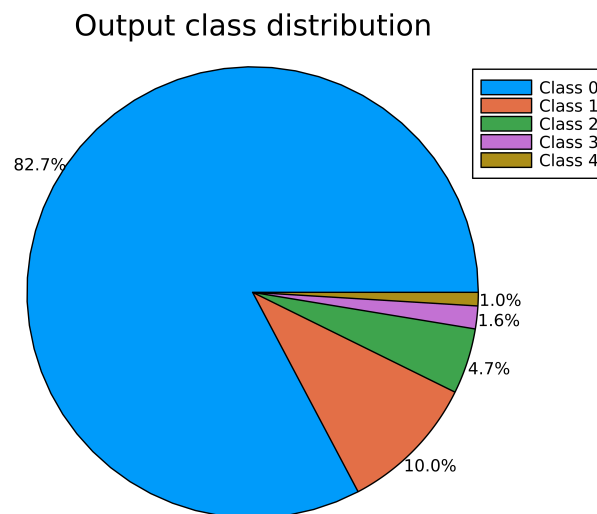


Fig. 1. Health Impact Class Distribution

This chart shows how the data is distributed among the different health impact classes. It provides a visual representation of the class balance in the dataset, which is important for understanding the model's training and performance.

In conclusion, the dataset provides comprehensive data on air quality and health outcomes, making it an excellent resource for building models that classify health impacts based on environmental data.

4 Methodology

In this section, we will discuss various approaches for processing the dataset. Each approach includes important elements such as dataset description, data preprocessing, methodology, and experimental results.

Table 1. Normalization Comparison

Approach	Accuracy	Error Rate	F1-Score
MinMax	93.89%	6.11%	2.968
ZeroMean	92.18%	7.82%	2.339

Table 2. Cross Validation

Approach	Accuracy	Error Rate	F1-Score
CrossVal	94.15%	5.85%	2.662

4.1 Data Preprocessing

Data preprocessing is a crucial step in any machine learning project. In this experiment, two normalization techniques were compared during the initial stages: *min-max normalization* and *zero-mean normalization*. Based on the results, we observed that min-max normalization consistently produced better outcomes with cross validation and was subsequently adopted in approaches 3 and 4.

Min-max normalization scales the data to a fixed range, typically between 0 and 1, which offers several advantages:

- It helps in handling the varying scales of different features, especially when features have different units.
- It ensures that no feature dominates due to its larger scale.
- It works well with many machine learning algorithms, including the ones we are using in this project.

The normalization was performed using the minimum and maximum values of each feature in the dataset, ensuring proper scaling and uniformity across all features.

4.2 Cross-Validation for Model Evaluation

We performed cross-validation on the normalized data to assess the performance of the models. We used a 10-fold cross-validation approach, meaning the dataset was split into 10 parts, with each part used for testing while the rest were used for training. This approach helps ensure that our model generalizes well and does not overfit to specific data points.

Cross-validation is particularly useful in ensuring the robustness and reliability of the model's performance across multiple data splits. This technique also helps in identifying potential issues like overfitting or underfitting during model training.

4.3 Ensemble Method

An ensemble technique was applied to improve the performance of the individual classifiers. This involved using a stacking method to combine the predictions from the models.

To construct the stack, the best-performing classifiers were selected based on their accuracy during initial evaluations. These included:

- The best *Artificial Neural Network (ANN)*,
- The best *k-Nearest Neighbors (k-NN)*,
- The best *Support Vector Machine (SVM)*, and
- The best *Decision Tree (DT)*.

Table 3. PCA Comparison

Approach	Accuracy	Error Rate	F1-Score
PCA_3	83.73%	16.27%	1.271
PCA_095	93.59%	6.41%	2.65548

The *Stacking Ensemble* method was chosen for its ability to combine the strengths of *Hard Voting* and *Soft Voting* techniques, resulting in a more robust and flexible model.

4.4 Application of PCA

We explored two approaches for dimensionality reduction using Principal Component Analysis (PCA). The first approach reduced the dataset from thirteen to three features, while the second preserved 95% of the data's variance.

The first approach yielded poor results, suggesting that significant information essential for effective training was lost during the dimensionality reduction process.

In contrast, retaining 95% of the variance allowed PCA to reduce the dataset's dimensionality effectively. This approach improved model efficiency, reduced overfitting, and delivered faster training times while maintaining strong predictive performance.

To illustrate the impact of these approaches, the table below compares the accuracy, error rate, and F1-score for both approaches.

The results show that **PCA_095** (preserving 95% of the variance) outperforms **PCA_3**, which significantly reduces the dataset's dimensionality to only three features, losing crucial data for effective training.

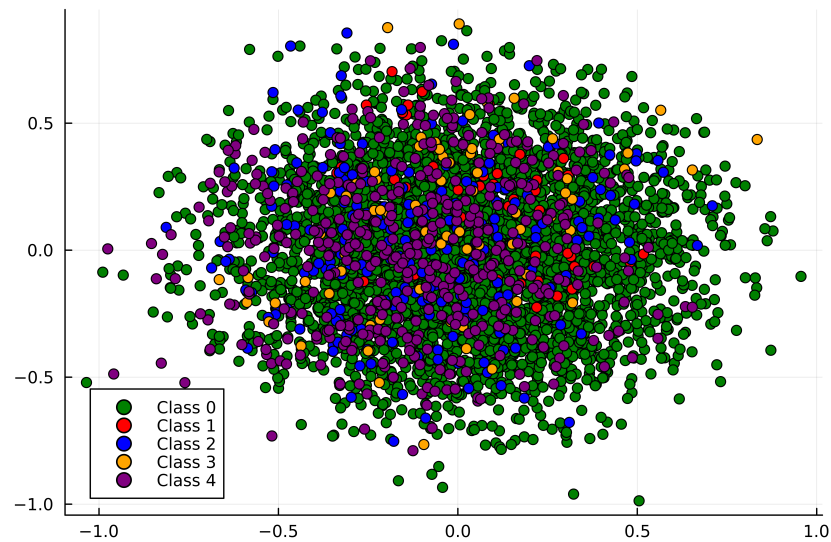


Fig. 2. PCA Results with 3 features

The figure above shows the results for PCA when reducing the data to 3 features. The reduced dimensionality resulted in a noticeable decrease in model performance, as shown by the lower accuracy and F1-score.

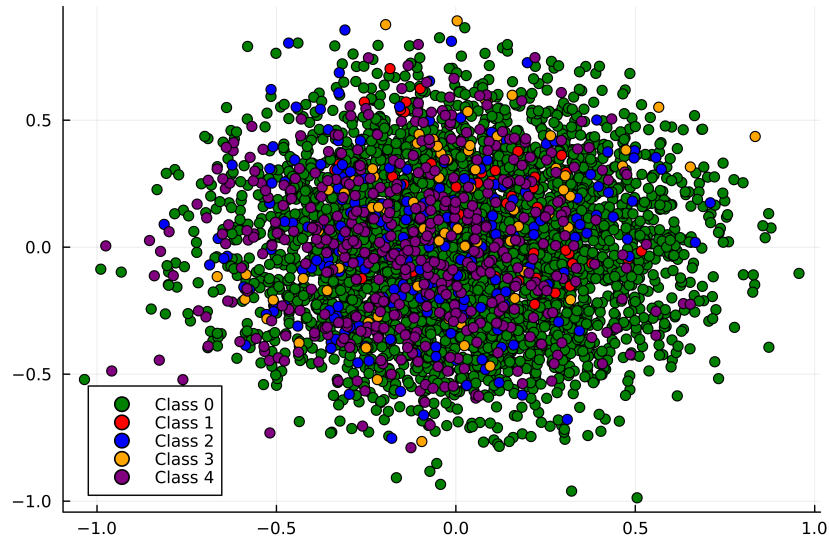


Fig. 3. PCA Results preserving 95% of data variance

Table 4. Model Performance Comparison: Accuracy, Error Rate, and F1-score

Approach	Accuracy	Error Rate	F1-score
MinMax	0.938951	0.061049	2.96801
ZeroMean	0.921754	0.0782459	2.3399
CrossVal	0.94148	0.0585196	2.66209
PCA_3	0.837349	0.162651	1.271
PCA_095	0.935886	0.0641145	2.65548

On the other hand, when PCA preserved 95% of the data's variance (as shown in the figure above), the results were much more promising, with significantly better accuracy and performance. This method strikes a balance between reducing dimensionality and maintaining critical information.

In conclusion, **PCA_095** is the preferred dimensionality reduction approach for this dataset, as it provides a much better balance between efficiency and predictive performance.

4.5 Reporting Experiments with Metrics and Discussion

Each experiment was evaluated using performance metrics such as accuracy, error rate, and F1-score. We compared the models based on these metrics and discussed the results in detail. This helped in understanding which model performed the best under different conditions.

The following table summarizes the performance of each model with respect to accuracy, error rate, and F1-score:

From the table above, we can observe the following:

- **Accuracy:** This metric indicates how often the model's predictions were correct. The **CrossVal** approach achieved the highest accuracy (94.15%), closely followed by **PCA_095** (93.59%). On the other hand, **PCA_3**

performed the worst with an accuracy of 83.73%, which suggests that reducing the number of features to just 3 may have lost important information.

- **Error Rate:** The error rate measures the proportion of incorrect predictions. As expected, the models with higher accuracy also have lower error rates. **CrossVal** (0.0585) and **MinMax** (0.0610) showed the best performance in terms of minimizing errors, while **PCA_3** had the highest error rate (0.1627), confirming that reducing features too much can negatively affect model performance.

- **F1-score:** The F1-score combines precision and recall into a single metric, which is especially useful for imbalanced datasets. The **CrossVal** approach achieved the highest F1-score (2.66209), followed closely by **PCA_095** (2.65548). **PCA_3** had the lowest F1-score (1.271), suggesting that the model's ability to balance precision and recall suffered when too many features were removed.

In conclusion, **CrossVal** performed the best across all three metrics (accuracy, error rate, and F1-score), while **PCA_3** underperformed due to excessive feature reduction. The **MinMax** and **PCA_095** approaches provided good overall performance, with slight variations in the F1-score.

4.6 Justification of the Metrics

The metrics used in this project are **accuracy**, **error rate**, and **F-Score**, as they provide essential insights into the model's performance.

- **Accuracy** measures how often the model makes correct predictions. It gives an overall view of the model's effectiveness and helps determine if the model is working well in general.
- **Error rate** indicates the frequency of incorrect predictions. It highlights the areas where the model is making mistakes, with a lower error rate reflecting fewer errors.
- **F-Score** balances *precision* (the accuracy of positive predictions) and *sensitivity* (the ability to correctly identify positive instances). This metric is valuable in ensuring the model is both accurate and comprehensive in its predictions.

Additional metrics such as **sensitivity**, **specificity**, **positive predictive value**, and **negative predictive value** are also considered. These metrics provide further understanding of how well the model identifies both high-risk and low-risk cases, helping to minimize issues like false positives (incorrectly identifying a non-risk case as a risk) and false negatives (failing to identify a true risk case).

Using these metrics together offers a well-rounded evaluation of the model's performance and highlights areas for potential improvement.

4.7 Supporting Material

We have included confusion matrices and explanatory plots to further support the claims made about the models' performance. These visual aids help to understand how well each model performed in predicting the correct classes, as well as identifying the types of errors the model made.

Confusion matrices are a powerful tool for evaluating classification models because they provide a detailed summary of how the model is classifying each class. Each cell in the matrix represents the number of predictions made by the model for a specific class. For binary classification problems, the matrix consists of:

- **True Positives (TP):** The number of times the model correctly predicted the positive class.
- **False Positives (FP):** The number of times the model incorrectly predicted the positive class when it was actually negative.
- **True Negatives (TN):** The number of times the model correctly predicted the negative class.
- **False Negatives (FN):** The number of times the model incorrectly predicted the negative class when it was actually positive.

From the confusion matrix, we can derive important metrics such as:

- **Accuracy:** The percentage of correct predictions out of all predictions. Accuracy gives us an overall sense of how well the model is performing.
- **Error Rate:** The proportion of incorrect predictions. It is calculated as $1 - \text{Accuracy}$.
- **Precision and Recall:** These metrics are derived from the confusion matrix and provide more insights into the model's performance, especially in imbalanced datasets.
- **F1-Score:** The harmonic mean of precision and recall. It is a good indicator when we need to balance precision and recall.

We have generated confusion matrices for each of the approaches we tested (e.g., **MinMax**, **ZeroMean**, **PCA_3**, **PCA_095**, and others). Each confusion matrix includes these metrics to help us understand the model's strengths and weaknesses in predicting different classes.

The following figure presents the confusion matrices for each approach, along with key metrics such as accuracy, error rate, and F1-score:

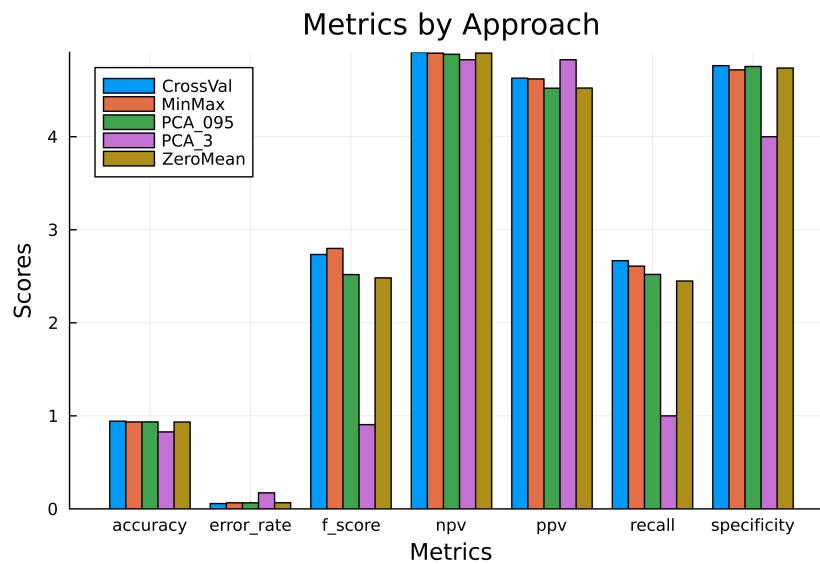


Fig. 4. Confusion Matrices for Each Approach

By analyzing the confusion matrices and the associated metrics, we can clearly see which models are performing well and which are struggling. This detailed analysis allows us to make better decisions about which models to deploy in practice, depending on the specific requirements of the application (e.g., minimizing false positives or maximizing accuracy).

4.8 Justification of Choices Made for Our Approach

For our project, we chose the following approaches for the reasons mentioned below:

- **Min-Max Normalization:** This method scales the data to a fixed range, usually between 0 and 1, preventing features with larger values from dominating the model. It's particularly useful for algorithms like SVM and neural networks, which are sensitive to the scale of the data. Since our data doesn't have any outliers, Min-Max normalization worked well and gave good results. Outliers can sometimes affect Min-Max, but in our case, the data was clean, so it helped improve model

Table 5. Artificial Neural Network (ANN) Configurations

Name	Topology	Max Epochs	Learning Rate
ANN_64	(64)	200	0.01
ANN_128	(128)	500	0.005
ANN_64_32	(64, 32)	400	0.001
ANN_128_64	(128, 64)	200	0.01
ANN_256	(256)	300	0.05
ANN_128_64_32	(128, 64, 32)	200	0.01
ANN_64_64	(64, 64)	250	0.005
ANN_256_128	(256, 128)	200	0.001

performance. Compared to Zero-Mean normalization, which centers the data around zero, Min-Max was a better choice because it avoided any issues with data skewing due to outliers.

- **Cross-validation:** Cross-validation allows us to evaluate the model's performance on different subsets of the data, ensuring that the results are reliable and not influenced by overfitting.
- **Stacking Classifier:** Stacking allows us to combine the strengths of multiple models, improving overall performance by using the predictions from different models to make the final decision. This ensemble approach generally yields better results than using a single model.
- **PCA with 95% Variance:** PCA helped reduce the number of features while preserving most of the data's important information. This made the model faster and more efficient without sacrificing performance.

5 Results

For our experiments, we used the following techniques:

- **Artificial Neural Networks (ANNs):** We tested 8 different architectures with varying numbers of hidden layers. These architectures can be seen in Table 5. We experimented with different configurations of neurons and layers to observe how they affect the model's performance. The hyperparameters we tested include:
 - **Number of Hidden Layers:** We tested models with one or two hidden layers.
 - **Number of Neurons:** The number of neurons in each hidden layer (e.g., 64, 128, 256).
 - **Max Epochs:** The maximum number of training iterations, which we varied to ensure the model was trained sufficiently.
 - **Learning Rate:** Different learning rates (e.g., 0.01, 0.005) to adjust how fast the model learns.
- **Support Vector Machines (SVM):** We tested different kernels and values of the hyperparameter C , as shown in Table 6. The SVM configurations include:
 - **Kernels:** We tested RBF, Linear, Polynomial, and Sigmoid kernels, which define the decision boundary.
 - C : This parameter controls the trade-off between achieving a low error on the training data and having a smooth decision boundary. We tested values like 0.5, 1.0, and 10.0.
 - **Degree:** For Polynomial and Sigmoid kernels, we tested different degrees (e.g., 2, 3, 5) to adjust the curvature of the decision boundary.
- **Decision Trees (DT):** We tested 6 different depth values for the decision tree, as shown in Table 7. The hyperparameters we considered for Decision Trees include:
 - **Max Depth:** The depth of the tree, tested with values from 3 to 21, to control its complexity.
 - **Random State:** Ensures reproducibility of results.

Table 6. Support Vector Machine (SVM) Configurations

Name	Kernel	C	Degree
SVM_rbf_05	RBF	0.5	2
SVM_linear_1	Linear	1.0	4
SVM_poly_001	Polynomial	0.01	2
SVM_sigmoid_1	Sigmoid	1.0	3
SVM_rbf_001	RBF	0.01	3
SVM_poly_5	Polynomial	5.0	3
SVM_linear_05	Linear	0.5	3
SVM_sigmoid_10	Sigmoid	10.0	5

Table 7. Decision Tree (DT) Configurations

Name	Max Depth	Random State
DT_3	3	42
DT_5	5	42
DT_7	7	42
DT_11	11	42
DT_15	15	42
DT_21	21	42

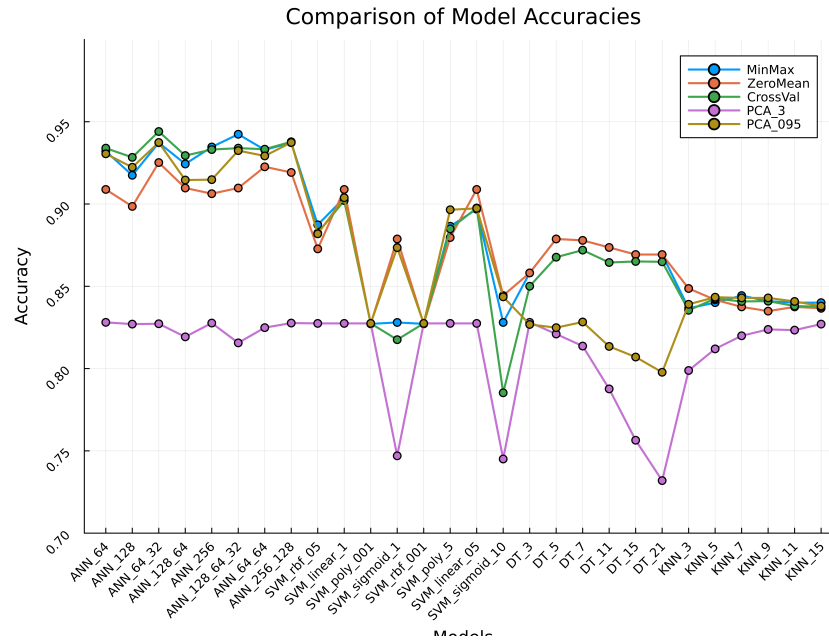
Table 8. k-Nearest Neighbors (kNN) Configurations

Name	Number of Neighbors (k)
KNN_3	3
KNN_5	5
KNN_7	7
KNN_9	9
KNN_11	11
KNN_15	15

- **k-Nearest Neighbors (kNN):** We tested six different values for k , the number of neighbors, as shown in Table 8. The hyperparameter for kNN includes:
 - **k:** The number of nearest neighbors considered for classification, ranging from 3 to 15.

6 Final Discussion

In this project, we evaluated and compared the performance of several machine learning models: **Artificial Neural Networks (ANNs)**, **k-Nearest Neighbors (k-NN)**, **Support Vector Machines (SVM)**, and **Decision Trees (DT)**. We tested these models using different preprocessing techniques and evaluated their performance based on key metrics: accuracy, error rate, and F1-score. The evaluation also included confusion matrices to understand how well each model classified the data and identify areas for improvement.



- **CrossVal**: Showed low misclassification rates and performed well across all classes.
- **PCA_3**: Had higher false positives and false negatives, indicating it struggled with classifying certain classes.
- **MinMax and PCA_095**: Despite their good performance overall, the confusion matrices revealed instances of misclassification, suggesting areas for further improvement.

6.4 Conclusion

- The **CrossVal** approach was the most effective, performing the best in terms of accuracy, error rate, and F1-score.
- The **stacking ensemble** method improved model performance by combining multiple models.
- **PCA_3**: Showed the drawbacks of excessive feature reduction, which can negatively affect model performance.
- **MinMax and PCA_095**: Performed reasonably well but did not outperform **CrossVal**.
- This study highlights the importance of choosing the right preprocessing method and model for classification tasks, and suggests that ensemble methods like stacking can lead to more robust and reliable predictions.

References

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